Praetor architecture

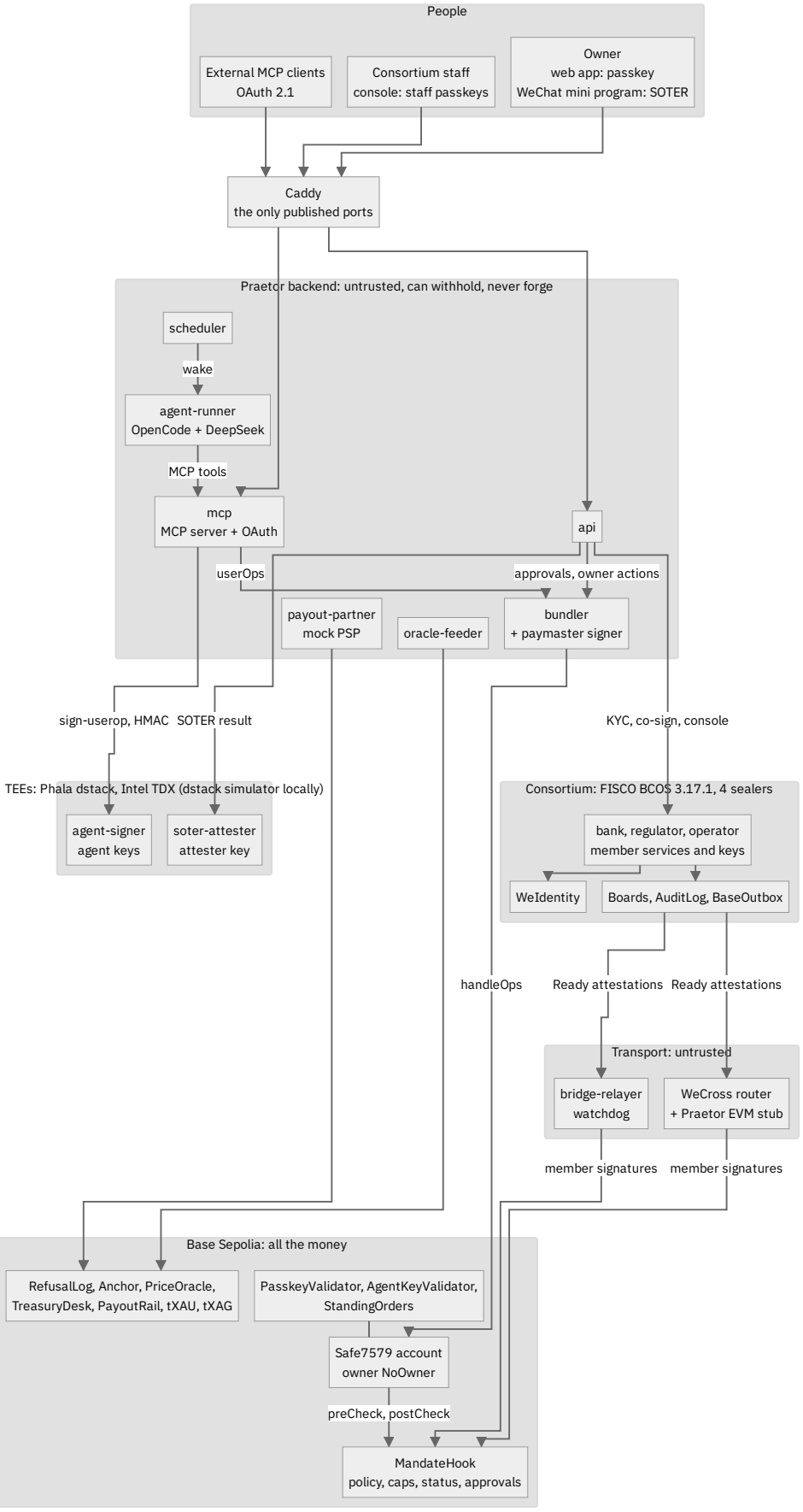
> **Self-driving money; the regulator holds the brakes.**

Praetor is an agent wallet (FinTechathon 2026, International Track, Topic F). An AI agent manages a user's portfolio (USDC, tokenized gold tXAU, tokenized silver tXAG) and pays their bills on its own, from an ERC-4337 smart account on **Base Sepolia**. Three properties hold at once:

- **Autonomous**: inside its mandate the agent rebalances and pays without asking anyone.
- **Bounded**: every call the account makes passes an on-chain **Mandate hook** that applies the user's rules and the bank's caps, default deny. Anything unusual needs the owner's Face ID (a passkey, or SOTER inside WeChat).
- **Regulated**: a **FISCO BCOS** consortium (bank, regulator, operator) holds identity, approved assets, sanctions, the agent's status and an append-only audit log. Its decisions reach Base as member signatures that the hook verifies itself. One regulator signature freezes the agent; loosening needs two of three.

This document is the technical documentation for judges: the picture first, then the core algorithms, the security design and the deployment. It describes the code as built on 2026-10-18, after the web app, the mini program, agent key rotation and recovery (F10), the Base Sepolia deployment and the production compose were added, and after the red team, the M18 rehearsal and the Pyth feed on live (D86 to D90), the teammates' phone-test fixes (D91, D93 to D95) and the M16 live run on Base Sepolia (D92, D96); the public server and its two Phala Cloud CVMs followed on 2026-10-20 (section 6, D106); every Phase B milestone gate has passed, and section 7 gives the status of each area (D85). Where the build differs from `spec/`, the entry in DECISIONS.md is cited. Chinese version: zh-CN/ARCHITECTURE.md. Security evidence: security/SELF_ASSESSMENT.md.

1. The system in one picture

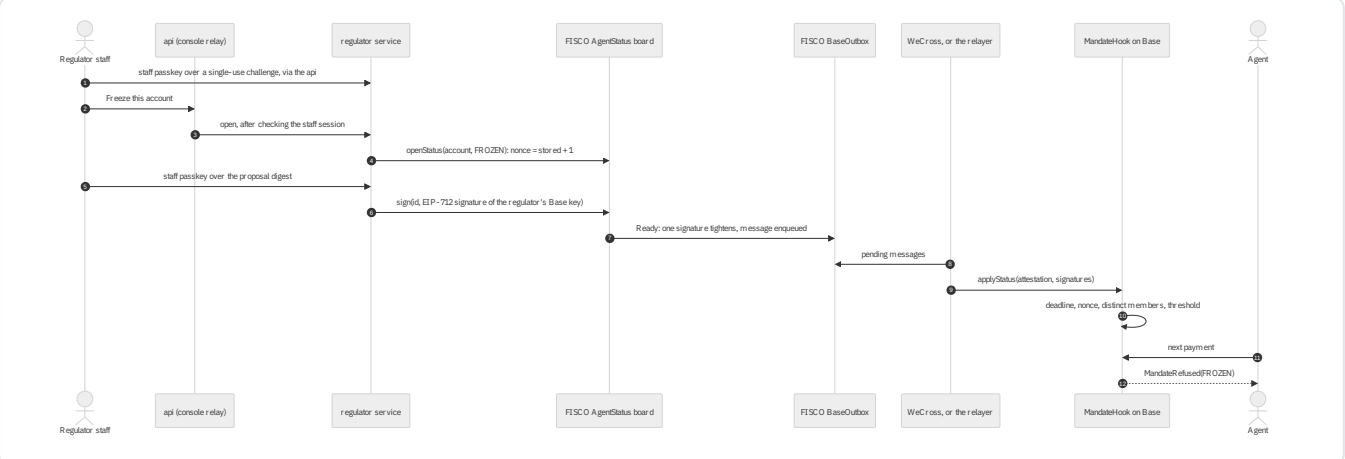


Trust boundaries. Trusted for safety: the hook's code, member keys within their thresholds, the owner's passkey, the TEE measurements the bank registered, Tencent's `verify_signature` answer, WebAuthn.sol with the P-256 precompile. Everything else (api, mcp, the LLM, scheduler, bundler, relayer, WeCross, oracle feeder within its bound) is untrusted: it can delay or withhold, it cannot forge a signature the hook accepts. Silence fails safe: without heartbeats the agent loses its autonomy.

Zone	Components (directory)
Base Sepolia (chain id 84532)	`contracts/base`: MandateHook with its libraries, PasskeyValidator, AgentKeyValidator, NoOwner, StandingOrders, PayoutRail, PriceOracle, TreasuryDesk, RefusalLog, Anchor, PraetorPaymaster, WeCrossInbox, the vendored WeCrossProxy, tXAU and tXAG (own ERC-3643 with IdentityRegistry and Compliance)
FISCO BCOS (chain id 20200)	`contracts/fisco`: MemberRegistry, AgentStatus, MandateRegistry, AssetRegistry, SanctionsList, JurisdictionRules, AttesterRegistry, AuditLog, BaseOutbox; WeIdentity contracts; WeCross hub and proxy
Member services	`services/bank`, `services/regulator`, `services/operator` on `packages/member-kit`
Backend	`services/api`, `mcp`, `agent-runner`, `scheduler`, `bundler`, `oracle-feeder`, `notifier`, `bridge-relayer`, `payout-partner`, `alipayplus-mock` (also the WeChat Pay mock), `kyc-vendor-mock`, `fake-llm` (tests only)
TEEs	`services/agent-signer`, `services/soter-attester` (dstack simulator locally, Phala Cloud CVMs in production)
Transport	`wecross/evm-stub` (Java 8), `infra/wecross`, `services/bridge-relayer`
Front ends	`apps/web` (React, the 3D phone, EN and 中文, PWA with web push, consortium console, recovery page, interim judges' sandbox); `apps/miniprogram` (WeChat mini program, native WXML, EN and 中文, SOTER approvals)
Shared	`packages/protocol` (EIP-712 types, reason codes, callHash, policy, audit chain, golden vectors in TS and Solidity), `chain`, `userop`, `config`, `db`, `fisco-client`, `weid-client`

2. Two chains and the bridge

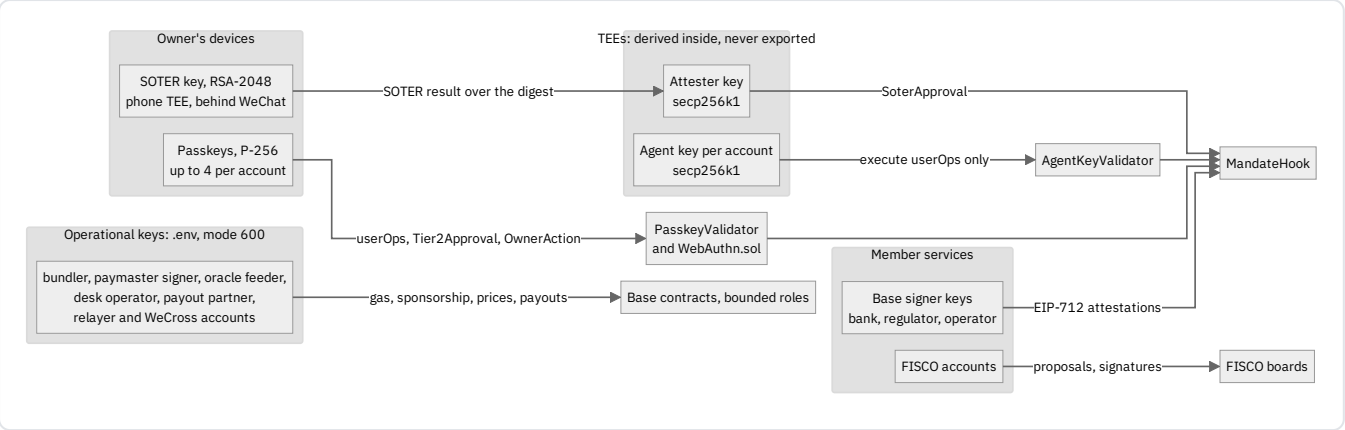
Money lives on Base; authority lives on FISCO. A consortium decision is a FISCO proposal that members sign with their Base signer keys; once it reaches its threshold it is queued for Base, where the hook checks every signature again. The transport cannot change the message, only delay it. The freeze of the demo, end to end:



The e2e test `test/e2e/F9-governance.test.ts` proves this flow (freeze applied on Base within 60 s, the next agent action refused FROZEN, loosening refused with one signature and accepted with two) and

`test/integration/wecross.test.ts` proves the WeCross path.

3. Who holds which key



Key	Can	Cannot
Owner passkey	Validate owner userOps (`execute`, and the launchpad's `setupSafe` for the first op); sign Tier2Approval and OwnerAction digests; sign in to the web app, only while PasskeyValidator holds it (D47)	Loosen anything without the bank (T3), including adding another device (`SET_OWNER_KEY`); produce an ERC-1271 signature
Agent key (agent-signer TEE, `getKey("praetor/agent/<account>/v<n>")`; a rotation moves to the next version, section 4.13)	Validate `execute` userOps of its account, to targets other than the account	Change rules, keys or modules; act outside the hook; sign for another chain or entry point (refused by agent-signer)
Attester key (soter-attester TEE)	Sign `SoterApproval` after Tencent verified a SOTER signature over the exact digest	Approve without a fresh SOTER signature; act once any one member deactivates it
Member Base signer keys	Sign consortium attestations; the bank's key also co-signs owner actions and alone signs a recovery's `SET_OWNER_KEY` after its re-KYC (section 4.13)	Exceed their threshold: 1 tightens, 2 loosen, the bank alone for caps, KYA, asset and attester activation
Member FISCO keys	Open and sign FISCO proposals, append audit entries (operator)	Anything on Base
Bundler, paymaster signer	Submit userOps and plain transactions to five allowlisted contracts; sponsor registered accounts	Forge a user or member signature
Oracle feeder	Push prices, at most 25 % per update	Trade
Desk operator, token agent (bank)	Register identities, operate the desk, freeze tokens; the ERC-3643 agent role can force-transfer tXAU and tXAG (known risk 12); send the bank's `scheduleRecovery` transaction (the authority is the bank signature it carries)	Move USDC; trade for others
Payout partner	Withdraw, settle or fail its own payouts on PayoutRail	Pull from wallets

All non-TEE keys live in `.env` (git-ignored, mode 600) in three sets: `LOCAL_*` (anvil fork), `LIVE_*` (Base Sepolia), `SANDBOX_*` (judges' sandbox). Services read the unprefixed name first, then the active set's prefix (D1). The EIP-712 domain name carries the environment (`Praetor`, `Praetor-local`, `Praetor-sandbox`), and each environment

has its own member keys, so a signature can never be replayed across environments even though the fork and live Base share chain id 84532.

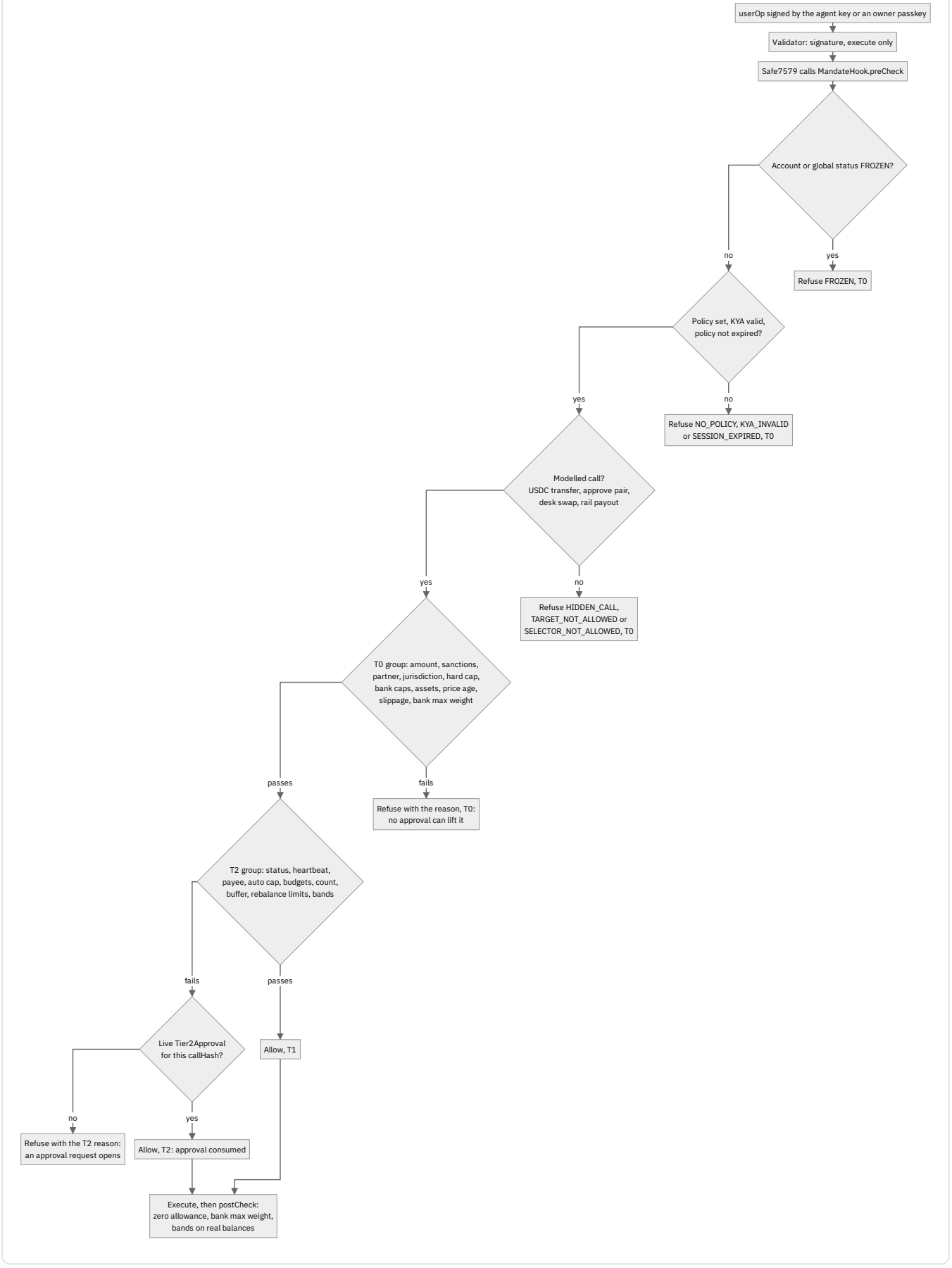
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4. Core algorithms

4.1 The MandateHook decision pipeline

`MandateHook` is the single global ERC-7579 hook of every account (Safe7579 supports exactly one). Safe7579 calls `preCheck` before and `postCheck` after every `execute`, `executeFromExecutor`, `installModule` and `uninstallModule`. The Safe's only native owner is `NoOwner`, whose ERC-1271 always fails, so `execTransaction` and the owner-signature fallbacks are dead ends (I1). Both validators return "invalid" to every ERC-1271 request, so USDC's `permit` and `transferWithAuthorization` cannot be used to pull funds around the hook (I16).

`preCheck` routes by selector: `execute` (single or batch mode only, canonical ABI encoding only), `executeFromExecutor` (only from StandingOrders), and module calls (only as the inner call of an approved module change, handed over in transient storage, D8). Everything else is `HIDDEN_CALL`. Each call of the execution then goes through one evaluator, `DecisionLib.evaluate`, which is also behind the views `check()` and `checkBatch()`. It never reverts on any input (I12): it returns a reason and a tier.



The checks, in the order the code applies them (spec 03, ported from BSA Sentry's `decide.rs`):

Step	Payment (`USDC.transfer`) and payout (`PayoutRail.pay` on Alipay+ or WeChat Pay)	Rebalance (`TreasuryDesk.swap`)	Tier
Base	FROZEN, NO_POLICY, KYA_INVALID, SESSION_EXPIRED; ETH value or a call to the account itself is HIDDEN_CALL	same	T0
Classify	exact selector and calldata length, canonical address padding; anything else on a known contract is SELECTOR_NOT_ALLOWED, any other target TARGET_NOT_ALLOWED	same	T0
T0 group	INVALID_AMOUNT (0 or above 2^128); SANCTIONED; payout partner not ACTIVE (COUNTERPARTY_NOT_APPROVED); JURISDICTION_BLOCKED; OVER_HARD_CAP above min(policy hard cap, bank per-transaction cap); BUDGET_EXCEEDED above the bank's daily or monthly cap	INVALID_AMOUNT; desk not ACTIVE; ASSET_NOT_APPROVED (one side USDC, bought token ACTIVE, sold token not DELISTED, both in the bands); STALE_PRICE; SLIPPAGE (`minOut` below oracle value minus `maxSlippageBps`); BAND_BREACH above the bank's max weight	T0
T2 group	RESTRICTED (status, owner pause or pending recovery); STALE_STATUS (heartbeat expired); UNKNOWN_PAYEE (payee or `merchantKey` not in the policy); OVER_AUTO_CAP; BUDGET_EXCEEDED (user budgets); COUNT_EXCEEDED; BUFFER_BREACH	RESTRICTED; STALE_STATUS; REBALANCE_LIMIT (`maxTradeUsd`, `maxRebalancesPerDay`); BAND_BREACH (an asset ends outside its band without moving closer to target, I15); BUFFER_BREACH	T2
Approve pattern	`approve(spender, amount)` is allowed only directly before the `swap` or `pay` that spends exactly that amount, and inherits its decision; `postCheck` requires the allowance back at zero	same	inherited
Effects	spend counters (day = `timestamp / 1 days`, month = `timestamp / 30 days`), count, approval consumed, one `Decision` event per call	rebalance counter, `Decision` event	

Running totals carry across the calls of a batch, so a batch cannot split a payment under a cap. A standing order executed by the StandingOrders executor applies the base and T0 checks and the approve pattern, but not the T2 group: the owner and the bank pre-approved that exact order. After execution `postCheck` reads real balances: the bought token must stay within the bank's max weight, and unless a human approved a band breach every band must end inside its range or closer to its target.

Refusals leave evidence. Safe7579 turns a hook revert into `ExecutionFailed()`, so the bundler submits refused ops anyway (rate-limited) and then calls `RefusalLog.log`, which recomputes the reason with `check()` instead of trusting the reporter (I11). The 28 reason codes of spec 02 are the wire values; the hook never returns `TIER2_REQUIRED` or `TIER2_INVALID`: an invalid approval makes its registration revert instead (D65).

4.2 Signatures: EIP-712 digests and passkeys

`packages/protocol` defines 13 EIP-712 types once, in TypeScript and in Solidity (`sol/PraetorTypes.sol`), and both sides are checked against the same golden vectors (`packages/protocol/vectors`). The domain is `{name, version "1", chainId 84532, verifyingContract = MandateHook}`. The chain id is the constant 84532 even on FISCO, so a FISCO board's `digestOf` is byte-identical to the digest the hook verifies (non-SM profile).

- **callHash** binds a human approval to one exact call: `keccak256(abi.encode(account, 84532, target, value, keccak256(data)))`.

- **Replay**: consortium attestations use `nonce == stored + 1` per key, allocated by FISCO (I18); Tier2Approval and OwnerAction carry a random nonce and the hook marks each digest used (`usedDigest`); expiry at most one hour; approvals are stored per account epoch, which recovery increments.
- **Passkeys**: `PasskeyValidator` keeps up to four P-256 keys per account, keyed by `credIdHash`, and verifies WebAuthn assertions with Coinbase's WebAuthn.sol (user verification required, high-s refused) on the RIP-7212 precompile at `0x100` (forge tests fall back to FreshCryptoLib). For a userOp the challenge is the userOpHash; for an approval or owner action it is the EIP-712 digest. The same check (`OwnerLib._ownerAuth`) runs inside the hook, so the owner's consent is verified on-chain, not by our servers.
- **Owner auth encoding**: `OwnerAuth{kind, data}` with `kind` PASSKEY (credential hash and WebAuthn data) or SOTER (`SoterApproval` and the attester's signature).

4.3 SOTER approvals through the attester TEE

SOTER is WeChat's biometric signature: the phone's TEE signs a JSON (`resultJSON`) that contains our challenge, and only Tencent can verify it (the public key is not exposed). Praetor turns it into an on-chain approval:

1. The mini program runs SOTER with the challenge set to the EIP-712 digest in hex, and sends the result with a fresh `wx.login` code.
2. The api forwards it to **soter-attester** (per-caller HMAC, D32). The attester exchanges the code for the openid, asks Tencent's `verify_signature` (must answer `is_ok`), and refuses unless `resultJSON.raw` equals the digest exactly.
3. It computes `bindingHash = keccak256(abi.encode(keccak(appId), keccak(openid), keccak(uid), keccak(cpu_id)))` (raw identifiers never reach a chain) and signs `SoterApproval{account, digest, bindingHash, soterCounter, expiresAt}` with its TEE key.
4. The hook accepts it only if the signer is an **active** attester (registered by the bank with the measurement of a verified quote, D33), account and digest match, the binding equals the account's linked binding, the expiry is within one hour, and the counter is above the stored one (`soterCounterStrict`, known risk 16).

The binding is set by `LINK_WECHAT`, an owner action that needs the passkey and the bank and carries a SOTER proof over the link digest (flow F2). SOTER is accepted only when the policy enables it, and never for `LINK_WECHAT` or `SET_OWNER_KEY`.

4.4 Consortium attestations: thresholds and nonces

On FISCO each decision type has a **board** contract (`contracts/fisco/src/Board.sol`): a member's FISCO account opens a proposal; the board allocates the proposal id and `nonce = stored + 1` for that key (one open proposal per key); members sign the EIP-712 digest with their Base signer keys; signatures are kept sorted by signer; at the threshold the board emits `Ready` and enqueues the attestation in `BaseOutbox`. An expired proposal is closed and releases its nonce; a Ready message that missed its Base deadline undelivered is reissued with the same nonce (D25). Member services sign only with a staff passkey assertion over the exact digest, except four automated, rule-checked signatures: the operator's heartbeat, the bank's KYA, the bank's owner-action co-signature and the bank's anchor co-signature (D24, D41). Outside the boards, the bank also signs a recovery's `SET_OWNER_KEY` alone, after its own re-KYC matched (section 4.13).

On Base, `ConsortiumLib` (a library of the hook) accepts a message only if `block.timestamp <= deadline`, `nonce == stored + 1`, and the signatures recover to distinct members in ascending order (no duplicates, no malleable signatures). Thresholds as built:

Message	Tighten	Loosen
Status (per account, or global with account 0)	1 member, when the new status is stricter	2 members (also for the same status)
Sanctions	1 lists	2 delist

Message	Tighten	Loosen
Jurisdiction (stablecoin payouts blocked)	1 blocks	2 unblock
Asset (token, desk, payout partner)	any 1 suspends or delists	the bank (activation, higher max weight, kind change)
Attester	any 1 deactivates	the bank activates
Caps per risk profile, KYA	the bank	the bank
Heartbeat	any 1, epoch = stored + 1, valid at most one day	
Anchor (`Anchor.anchor`)	any 2 members, seq = stored + 1	

The effective status of an account is the stricter of its own status and the global one. A freeze also blocks module changes, including removing the hook (D19).

4.5 Heartbeat fail-safe

The operator service opens a `HeartbeatAttestation` every `HEARTBEAT_INTERVAL_S` (60 s locally, 300 s in production) valid for `HEARTBEAT_TTL_S` (180 s locally, 900 s in production) after the latest Base block time, and signs it itself. If the heartbeat lapses (the consortium is down, the transport is cut, or someone withholds messages), every autonomous action is refused `STALE_STATUS`. It is a T2 reason: the owner can still approve a payment by hand, and the agent regains its autonomy at the next heartbeat. Withholding therefore never extends the agent's power.

4.6 Transport: WeCross, with the relayer as watchdog

`BRIDGE=wecross` is the local default (D42). The bridge-relayer, holding the WeCross router's FISCO account key, asks `BaseOutbox.requestWeCross(id)`; the outbox calls the WeCross hub; the **WeCross router** (1.4.0, stock BCOS3 stub on the FISCO side) polls the hub and hands the call to **Praetor's EVM stub** (`wecross/evm-stub`, Java 8, signing, RLP and ABI checked byte for byte against viem vectors), which sends a transaction to the vendored `WeCrossProxy` on Base. The proxy routes the name `inbox` to `WeCrossInbox.deliver(["Status", hex(abi.encode(attestation, signatures))])`, which calls the hook's permissionless setter and returns `["ok", nonce]` or `["error", reason]` without ever reverting the router's call. WeCross holds only transport keys; no member key enters it.

The relayer is the watchdog: any message WeCross has not applied within `BRIDGE_WATCHDOG_S` (120 s) it submits itself through the bundler, and it records deliveries in the outbox (first write wins). The delay is measured on FISCO's clock from the message's enqueue time (FISCO's web3 RPC reports block time in seconds, its EVM `block.timestamp` in milliseconds), so a request stuck in the WeCross hub is delivered whatever the Base clock says (`services/bridge-relayer/test/relayer.test.ts`). With `BRIDGE=relayer` it delivers directly. As a last resort any member can submit an attestation to the hook. WeCross's known issues and how they are contained: security/wecross.md.

4.7 Audit chain, Base anchors and the camt.053 statement

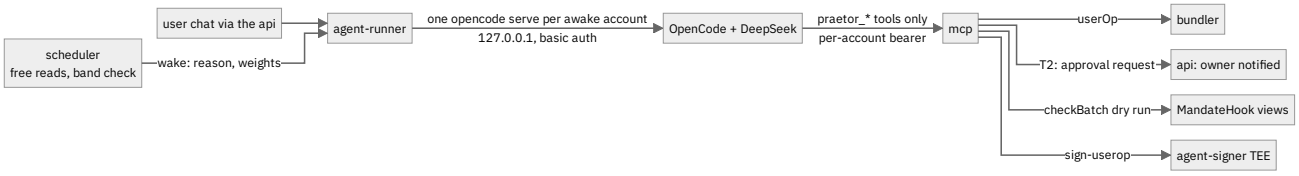
The operator's audit writer turns events into entries of an append-only hash chain: Base events of the hook, RefusalLog, StandingOrders, PayoutRail, TreasuryDesk and Anchor, and every FISCO `Ready` (through the outbox's `Enqueued`). Each entry is canonical JSON (RFC 8785, amounts as decimal strings, hex lowercased) with one of 18 kinds (DECISION, REFUSAL, APPROVAL, OWNER_ACTION, STATUS, HEARTBEAT, KYA, ASSET, SANCTIONS, JURISDICTION, CAPS, STANDING_ORDER, PAYOUT_REQUESTED, PAYOUT_SETTLED, REBALANCE, ANCHOR, KEY_ROTATION, RECOVERY), and

```
entryHash = keccak256(canonical JSON without entryHash)
head_n    = keccak256(abi.encode(head_{n-1}, entryHash_n)),  head_0 = 0x0
```

The operator stores the entry, appends its hash to `AuditLog` on FISCO (operator only), then marks it appended, so a crash never appends twice (D41). Every `ANCHOR_EVERY` entries (20) or `ANCHOR_INTERVAL_S` (one hour in production) the operator opens an anchor: `AuditLog.openAnchor` takes the sequence number, head and block from the log itself, so nobody can anchor a head that is not on the log. The operator signs; the bank co-signs only after recomputing the whole chain from the operator's JSON and reaching that head; two signatures make it Ready and it lands in `Anchor` on Base. An anchor left open (its co-signature lost in a restart) is resumed rather than opened again, which the board would refuse.

Verification (`/api/audit/verify`, web page Audit) uses three independent sources: it recomputes every hash from the JSON alone, compares the result with the FISCO heads (and bisects `headAt` to name the first tampered entry, even if its stored hashes were rewritten), and checks that the latest Base anchor is one of the recomputed heads. The **camt.053** bank statement (ISO 20022 `camt.053.001.02`) is built from the account's USDC transfers and payouts and validated against the official XSD with xmllint-wasm before it is returned; an invalid document is never served.

4.8 The agent runtime and the MCP server



- **Wake-ups.** The scheduler holds no key. Every `PRICE_CHECK_INTERVAL_S` (and at once on a DEMO price event) it reads each account's weights and wakes the agent when an asset is outside its band by more than 10 bps and rebalances are left today, at most once per account and breach within the cooldown (D36). Monitoring costs nothing; the LLM runs only on a wake or a chat.
- **Hosting.** Praetor hosts every user's agent: agent-runner starts one `opencode serve` per awake account (at most `AGENT_MAX_ACTIVE` = 8, least recently used idle instance evicted, 30 minutes idle stop). The model is `opencode/deepseek-v4.1-flash-free` (from `AGENT_MODEL`).
- **Tool lockdown** (D15). OpenCode's configuration is `permission: {"*": "ask", "praetor_*": "allow"}` and the runner rejects every permission request during a turn, so built-in tools (shell, files, web) never run. `webfetch`, `websearch`, `glob`, `grep`, `lsp` and `question` are denied in the configuration, so OpenCode does not offer them, and a turn whose pending permissions cannot be listed stops at once and aborts its session (D86, D88). The per-account MCP bearer reaches OpenCode only through an environment substitution, never in a file. agent-runner sits on its own `agent` network.
- **MCP server.** Ten tools: `get_portfolio`, `get_mandate`, `get_status`, `get_activity`, `explain`, `pay`, `rebalance`, `get_approval`, `propose_standing_order`, `trigger_standing_order`. **No tool takes an account**: the account comes from the bearer. `pay` and `rebalance` dry-run with `checkBatch`, then always submit, so a refusal is recorded on-chain as evidence; a T2 result opens an approval request for the owner. No tool reads KYC data (I9).
- **Bring any agent.** External MCP clients (for example claude.ai) connect through OAuth 2.1 with dynamic client registration and PKCE S256 (code ported from BSA Sentry); the user consents with their Praetor session. Their actions still go through the agent key in the TEE and the hook.
- **Signing.** agent-signer signs only for mcp, only `execute` userOps on chain 84532 and EntryPoint v0.7, never to the account itself, at most 120 per account per hour; it computes the userOpHash itself. mcp names the account's agent key as Base holds it (`kyaOf`) and agent-signer finds its key version among at most 16, so mcp never learns versions and a rotation needs no change on its side (D47).

- **After an approval.** Once the owner approved a T2 call, the api asks mcp to resubmit it; mcp signs only a single call whose approval is live and unconsumed on the hook (`approvalOf`), anything else is refused (D72).

4.9 Standing orders and payout rails

Fixed recurring payments (rent, subscriptions) are **standing orders** stored on-chain in the wallet's own `StandingOrders` executor: payee, amount, period, next due date, rail and a merchant hash. Only owner + bank create one (`ADD_STANDING_ORDER`, T3); the owner cancels alone. Anyone may trigger a due order; it pays exactly the amount, once per period, missed periods skipped; the hook still applies freeze, sanctions, jurisdiction, partner status and bank caps. On-chain because the wallet itself must enforce the authorisation: kept in a database, a compromised backend could change the amount or the payee. The agent may only propose an order (an owner request, D40).

Rails: `USDC_DIRECT` (a plain transfer) or Alipay+ / WeChat Pay through `PayoutRail`, an escrow: the account pays USDC in with `approve` + `pay(partner, amount, rail, merchantRef, paymentRef)`, and only the named partner withdraws, settles or fails it (a failure refunds the account). The payout partner (a mock PSP, labelled) runs the official `ant-intl-alipayplus-mcp` client against our Alipay+ mock gateway (RSA-signed requests, HTTPS inside the network) or the WeChat Pay mock. No personal data on Base: the merchant is `merchantRef = keccak256(merchantId)`; names and labels stay off-chain (I19).

4.10 Account creation, bundler and paymaster

The account is a Safe7579 created through its launchpad (addresses of spec 00): owners `[NoOwner]`, threshold 1, validators `PasskeyValidator(credIdHash, x, y)` and `AgentKeyValidator(empty)`. The first userOp, validated by the passkey, runs the init batch `[setRegistry(0, [], 0), installModule(2, StandingOrders), installModule(4, MandateHook)]` (the executor before the hook, D9). Validators are selected by the nonce key (the validator address in its top 20 bytes).

The **bundler** is ours (`POST /userop/prepare` then `/userop/send`, D22). It simulates every op and never submits one that fails validation (a failed first op bricks the address, Q20). `PraetorPaymaster` sponsors only registered accounts or Praetor's own launchpad init, so users never hold ETH (D20, D21). Refused ops are submitted for on-chain evidence, at most 20 per account per day. Plain transactions (consortium setters, anchors, standing orders, refusal logs) go only to five allowlisted contracts.

4.11 Identity: WeIdentity, KYC and KYA

- **KYC** at the bank: the Demo KYC provider (a labelled mock: document, selfie, liveness, sanctions name screening; kycLevel 2) or the EU Digital Identity Wallet (OpenID4VP, discloses only the birthdate and the country; kycLevel 1, D30). Personal data is sealed with AES-256-GCM in the bank's database and never leaves it (I9).
- **WeIdentity** on the same FISCO chain: the bank is an Authority Issuer (`KYC_PROVIDER`); it issues a `PersonKYC` credential to the user's WeID and an `AgentMandate` that the user approves with the passkey registered on the account and the bank countersigns. Verification is fail-closed: signature, evidence by the pinned issuer, revocation read without error (D28). WeIdentity's REST service is custodial and unauthenticated, so it lives on its own network (D29).
- **Agent identity**: the agent's WeID is created from the agent's public key, with services `AgentWallet = eip155:84532:<account>` and `MCP = <public URL>/mcp`, signed inside the TEE (D32).
- **KYA** (know your agent): the bank verifies the agent-signer's quote (key bound to a bank nonce, measurement on the allowlist; in production Phala's verifier checks the TDX quote and the bank reads the compose hash from the verified quote's MRCONFIGID, where dstack writes it, D106), then builds a `KYAAttestation` only from its own records: agent key, kycLevel, risk profile, jurisdiction, credential hashes, agent measurement (D37). Only after it reaches Base can the owner set that key as the agent key (`SET_AGENT_KEY`, owner + bank).

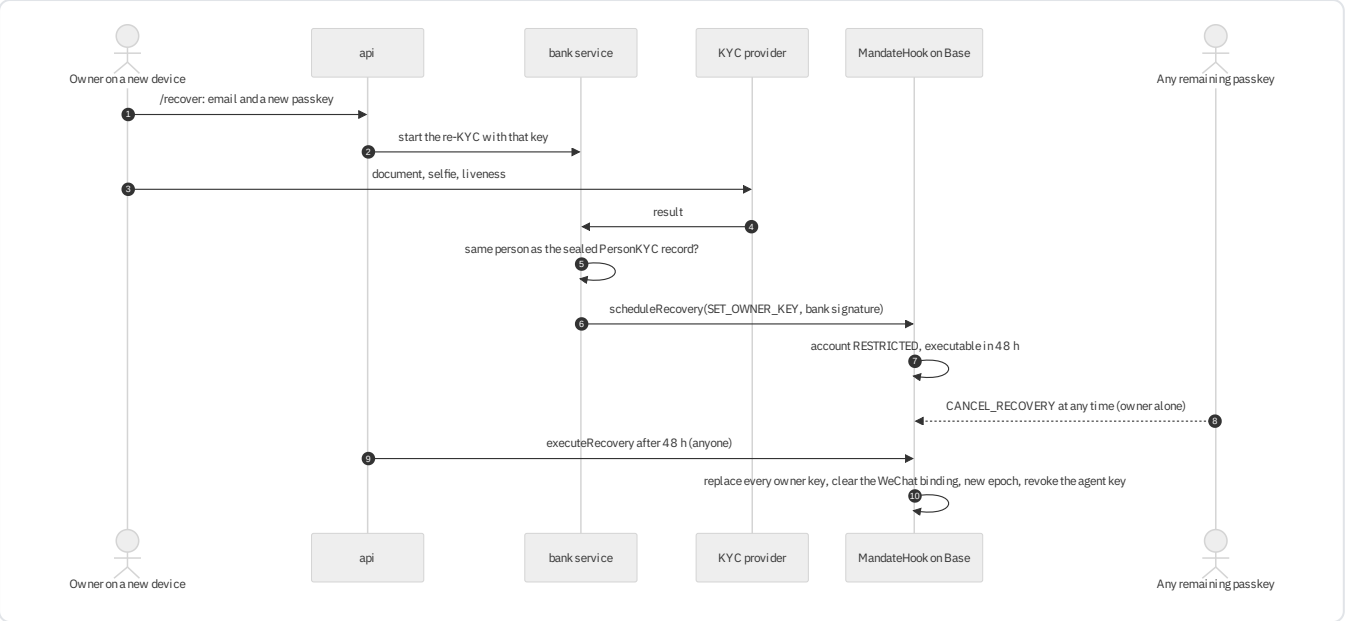
4.12 Prices and the treasury desk

`PriceOracle` holds 8-decimal USD prices for tXAU and tXAG (USDC pinned at 1). The oracle-feeder pushes Pyth Hermes prices only when `PYTH_API_KEY` is set and `CHAIN_MODE=live` (one request for both feeds every 5 minutes, D89); otherwise, and always on the fork, it pushes seeded prices flagged `DEMO` (D13); every update may move at most 25 %, and the hook refuses trades on prices older than `maxPriceAge` (4 days, metals do not trade at weekends). The demo shock (`pnpm shock --asset XAU --pct 20`) is a DEMO price the feeder holds for `DEMO_HOLD_S`. `TreasuryDesk` swaps USDC and metal at the oracle price plus or minus a 30 bps spread, minting what it sells and burning what it buys back. tXAU and tXAG are our own ERC-3643 tokens: transfers only between verified identities in allowed countries.

4.13 Agent key rotation, devices and owner recovery (F10)

The chain stays the source of truth for keys; the api only prepares what the owner and the bank sign (D47).

- **Stop my agent.** `REVOKE_AGENT_KEY` is an owner-alone action (T3-): the key stops validating at once. The api then asks the bank to revoke the AgentMandate evidence of every agent key that AgentKeyValidator no longer holds; the bank reads Base itself before revoking.
- **Rotate.** A new agent key is the next agent-signer key version (`praetor/agent/<account>/v<n+1>`). `POST /api/agent/rotate` finds the version whose address Base holds (`kyaOf`, at most 16 versions) and runs the onboarding path for the next one: quote checked by the bank, agent DID, AgentMandate signed with the passkey, a KYA that the bank links to its predecessor on FISCO (`MandateRegistry.linkSuccessor`), then `SET_AGENT_KEY` (owner and bank). The old key stops validating and its mandate is revoked as above.
- **Devices.** Settings registers another passkey (at most four per account) and adds it on Base with `SET_OWNER_KEY`, signed by a current passkey and co-signed by the bank. Sign-in accepts only passkeys that PasskeyValidator holds, so a new device signs in once Base has it, and an owner action, approval or mandate is signed as the credential named in the assertion's id.
- **Recovery** when every passkey is lost: the user opens `/recover`, enters the email and creates a new passkey; the bank runs its own re-KYC with the KYC provider, and only a result matching the sealed PersonKYC record (name, birth date, nationality, document number) makes it sign the `SET_OWNER_KEY` alone and send `scheduleRecovery` from its desk account. The account is RESTRICTED for 48 h, any remaining passkey cancels alone (`CANCEL_RECOVERY`), and after 48 h anyone may execute (the api through the bundler). Recovery needs a PersonKYC issued from the provider (not the EUDI path). The recovery state comes from the hook's Recovery events; Settings shows a pending recovery, the notifier pushes every `RecoveryScheduled` to the owner's devices (D81) and the bank's own notice goes to its log sink (D13). An email resolves to its first account; the new passkey is stored only once the bank has started its re-KYC, at most five a user an hour, and the recovery options name none of the account's passkeys (D80).



After execution the old approvals are void (the epoch changed) and the owner installs a new agent key. The flow is covered by `test/e2e/F10-rotation-recovery.test.ts` and, because it moves the fork clock 48 h, by `test/warp/F10-recovery-execute.test.ts`, which `make e2e` runs last; both passed at the M15 gate on fresh forks. Its risks are in the self-assessment, known risk 31.

4.14 The web app and the consortium console

`apps/web` (React and Vite, EN and 中文, an installable PWA) is the owner's main screen: `/` the landing page with the 3D phone, `/start` onboarding, `/home` portfolio rings and decisions, `/approve/:id` Tier-2 approvals, `/mandate`, `/payments` (standing orders, payees with the names the agent resolves), `/agent` (chat with the tools each turn called, stop or rotate the agent key), `/audit` (verify, camt.053), `/settings` (devices, a pending recovery, WeChat link and unlink, web push), `/recover`, `/sandbox` and `/console`.

- **What you see is what you sign** (D46). Before the passkey signs, the page recomputes the Tier2Approval digest from the request's preimage and, for every owner action, checks the kind it asked for, the account, `payloadHash` against the payload and the EIP-712 digest; for a policy, an agent key, a new passkey, a revocation and a recovery cancellation it rebuilds the payload from what it shows (`apps/web/src/lib/ownerAction.ts`). A lying api gets no signature.
- **Console.** One panel per member (bank, regulator, operator) side by side; each panel signs in with its own member service's staff passkey and asserts with that enrolled credential. A proposal opened from the console shows in every member's panel with a tracker of who signed and how many signatures it needs: the regulator freezes alone, loosening needs a second member. The api only relays; the member service verifies the staff assertion over the digest it reads from FISCO. The list of accounts needs a staff session (D70).
- **Browser controls** (D71). Caddy sends a CSP with scripts from self only and no framing, HSTS, nosniff and no-referrer; the api refuses a write from another Origin and rate-limits the public sign-in, onboarding and recovery routes per client.
- **Tests.** Unit tests in `make test` (`apps/web/src/lib/web.test.ts`, `apps/web/src/pages/Settings.test.tsx`); the Playwright journey F1 to F10 with a virtual authenticator, screenshots at 1440 and 390 px and axe checks (`apps/web/e2e`, `make e2e-web`), green at the M13 gate (15/15 with the sandbox spec, 34 screenshots in `docs/screens/web`, each with axe).

4.15 The WeChat mini program

`apps/miniprogram` is the owner's daily screen inside WeChat: portfolio, status, decisions, the agent chat and Tier-2 approvals with Face ID or fingerprint through SOTER (native WXML, WXSS and JS, EN and 中文).

- **Sign-in** (D50). `wx.login` gives a code; `POST /mp/session` lets the api call code2Session itself and keep only `keccak256(openid)`. The bearer token lasts 2 h, is never the web session, and names the account only while the link recorded after LINK_WECHAT still matches `bindingOf(account)` on Base, checked at every sign-in and every request (D81), so an unlink or a recovery (which clears the binding on Base) ends its sessions at once.
- **Routes** (D51, D53). `/mp` gives the linked user the owner's reads, the agent chat, reject, SOTER approvals and the F2 link, and no owner action: unlink, new standing orders and policy changes copy the web app link, where the passkey signs. No response carries the owner's email or KYC data.
- **What the phone checks** (D52, D54). Before SOTER signs an approval, the phone decodes each call, recomputes the call hash and the Tier2Approval digest and signs its own digest; for a link it requires LINK_WECHAT with `payloadHash = keccak256(abi.encode(linkNonce))`. The digest code is `lib/protocol.js`, generated from `src/protocol.mjs` and checked against the golden vectors.
- **Status.** Logic tests in `make test` (a stand-in for the DevTools build); on phones it runs in debug mode under the WeChat test account (MANUAL.md section 5, TESTER.md). On 2026-10-18 the team passed on an iPhone the link, a SOTER Face ID approval with the same digest on the phone and the web app, a replayed approval refused on-chain, the regulator freeze, the unlink and the Chinese UI; the Android fingerprint is not tested yet.

4.16 Notifications and web push

The notifier fans each message out to every channel of the account and keeps an outbox: the WeChat Official Account channel (a log sink until the OA test account exists, BLOCKERS B2) and web push. The web app subscribes its service worker from Settings with the VAPID public key of the api's public config; the api stores the subscription (`POST /api/push/subscribe`, browser push services only and at most ten per account, D82) and the notifier reads it through an internal route, encrypts the message for each browser (RFC 8291) and signs the request with Praetor's VAPID key (RFC 8292). The service worker shows the notification and opens its link when it is a page of the app (D82). A push service down never blocks the other channels.

4.17 The interim judges' sandbox

Spec 00's own sandbox deployment (domain `Praetor-sandbox`, SANDBOX keys) is not built; `/sandbox` is its interim stand-in on the main deployment (D48, D73). One click creates a disposable account with the browser's own passkey, passes the demo KYC, installs the agent and a default mandate, and funds it from the faucet: `POST /api/sandbox/faucet` mints 5,000 demo USDC once per account, at most `SANDBOX_RATE_LIMIT_PER_HOUR` per client, and only on the fork. The scenario buttons run real flows and show their transactions: a 300 USDC payment to a new café (real model, approved with the passkey), a prompt injection towards the demo sanctioned address, a rent standing order, the audit check. The freeze links to the console: no member key signs anything for a button. On the public server the sandbox is off (`/config` returns `sandbox: false`, and `/sandbox` leads to the onboarding at `/start`): judges run it on a local fork (`make up`).

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5. Security design

The full analysis, with the tests behind each claim, is security/SELF_ASSESSMENT.md. In short:

- **One choke point.** Every execution passes the hook, whoever signed it; Safe owners cannot execute; no signature path exists outside the hook (I1, I16, I17, proven by fork tests on the real Safe7579).
- **Default deny and bank caps first.** Only four call shapes are modelled; the effective limit is always the stricter of the user's policy and the bank's caps, and no approval lifts a T0 reason (I2, I8, fuzz and invariant tests).
- **Humans approve on-chain.** Approvals are bound to one exact call, single use, expiring within an hour, verified by WebAuthn.sol or an attested SOTER approval (I5, I6).

- **What you see is what you sign.** The web app and the mini program recompute every digest they sign from what they show, so a compromised api gets no signature for something else (D46, D52).
- **Stricter needs fewer signatures than looser.** One member freezes; two loosen; nonces allocated by FISCO cannot be jumped (I3, I18). Silence fails safe (I4).
- **Evidence by construction.** Refusal reasons are computed on-chain (I11); every decision lands in a hash chain anchored on Base.
- **Data minimisation.** KYC data stays in the bank; Base carries addresses, amounts and hashes plus the pseudonymous KYA attributes (I9, I19, D69).

The tiers in one line each: **T0** never (hook refuses); **T1** the agent alone; **T2** the owner by passkey or SOTER; **T3** owner and bank (owner alone to make things stricter); **T4** the consortium.

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6. Deployment

Local is a small production (spec 11): one Docker Compose file (`infra/compose.yml`) with profiles `core`, `chain`, `wecross`, `agent`, `tee-sim`, `mocks`, `dev`; every pulled image pinned by digest; healthchecks on every long-running service. Only Caddy publishes ports (80 and 443; anvil also listens on loopback 18545 for Foundry). Three networks isolate what must not talk: `praetor` (10.88.0.0/16), `identity` (WeIdentity with the bank, MySQL and one FISCO node only, D29) and `agent` (agent-runner with mcp, the scheduler, the api and the test-only fake LLM, D35). MySQL has one database and one user per service. Base runs as an anvil fork of Base Sepolia with the P-256 precompile (`--network optimism`), which retries the public RPC and dumps its state to a volume every 20 s (D45); FISCO runs four sealers from our own image built from the sha256-pinned 3.17.1 release; the TEEs run on the dstack simulator.

Live mode (M16, D74, D75). `make up LIVE=1` adds `infra/live.env`: the stack runs on Base Sepolia with the `LIVE_` keys, without anvil, against `infra/deployments/base-sepolia.json` with domain `Praetor` (for the M16 run, MandateHook `0x839E104c8C2df983213a89CAcB58440Da4AdC2A9`, deployed at block 48253270 and now kept in `infra/deployments/base-sepolia-m16.json`; since 2026-10-19 the file holds the server's redeployment, MandateHook `0x6411cc907Ace53F1Bdf8607a85991d3b38E79a5f`, block 48301428, D101), at `DEMO_SCALE=100` and production heartbeat and price intervals, with real XAU and XAG prices from Pyth Hermes (`FEED_INTERVAL_S=300`, D89). FISCO stays the local chain, governed by the `LOCAL` set (`FISCO_KEYSET=LOCAL`), with a second consortium bound to the M16 hook (`infra/deployments/fisco-live.json`, until the server commits its own); every key Base trusts comes from the `LIVE_` set. Live reads Base through `https://base-sepolia-rpc.publicnode.com` (`BASE_RPC_URL` in `infra/live.env`): `sepolia.base.org` caps `eth_getLogs` at 200 blocks and several services read wider ranges; publicnode allows 50,000 blocks, so every Base log read goes in windows of at most `BASE_LOGS_MAX_RANGE` blocks (10,000 by default, D98). The WeCross EVM stub sends its own User-Agent, because Cloudflare in front of the public RPCs refuses Java's default (D92). The live budget is small: `scripts/deploy/fund-live.ts` spreads the deployer's testnet ETH to the gas accounts (dry run unless `--send`), and one deployment with a 0.004 ETH paymaster deposit cost about 0.0045 ETH. `scripts/evidence/live-run.ts` runs the evidence flows with the real model and `scripts/evidence/collect.ts` writes the evidence from the chains. The evidence run of 2026-10-18 (M16, D96) is in EVIDENCE.md: C0 to F8 on one account with Pyth prices, the real model and WeCross, within the budget (D74). It also found that the scheduler and the oracle feeder must retry a failed tick within 30 s and that the payout partner must resume a payout interrupted between withdraw and settle; both are fixed (D96).

Production (deployed on 2026-10-20, D104, D106): one server for the stack, https://app.praetorwallet.com (Google Cloud `e2-standard-4` in Hong Kong, a Let's Encrypt certificate through Caddy), plus two Phala Cloud CVMs in real Intel TDX (dstack 0.5.12, production OS image) for agent-signer and soter-attester, with `ATTESTATION_MODE=phala`, compose hashes registered by the bank, secrets as encrypted environment variables, and the live Base Sepolia deployment of 2026-10-19 (D101) with the server's own FISCO consortium. The bank verified the attester's quote through Phala's verifier, reading the compose hash from the quote's MRCONFIGID, and activated it on Base; the app ids, compose hashes and the transactions of the first WeChat run are in EVIDENCE-

SERVER.md. Callers reach each CVM through Phala's gateway, which terminates TLS outside the CVM (TEE checklist item 6). `infra/compose.prod.yml` adds to the same base file a restart policy for every long-running service and log rotation for all, keeps the scripted model out (a `test` profile), lets Caddy take a real certificate (`CADDY_TLS` is the ACME account email instead of `internal`) and disables the plain-HTTP tunnel site (its host name never resolves). `make rehearsal` starts the stack with it (`PRAETOR_ENV=rehearsal`), runs the smoke subset of the flows, then `scripts/deploy/prod-refusals.sh`, which checks that each service refuses its test-only settings with `PRAETOR_ENV=prod` (I13); it passed on the fork on 2026-10-18 (D90). Phones reach a laptop stack before the domain exists through `make tunnel`: a cloudflared quick tunnel that reaches Caddy over plain HTTP, writes its URL to `.run/tunnel-url` and points the mini program's develop environment at it (TESTER.md). Steps: DEPLOY.md; TEE checklist: security/tee-checklist.md.

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7. Build status on 2026-10-18

"Built" means the code and its unit tests are in `make test`; a milestone gate is passed only when `docs/PROGRESS.md` ticks it (D85).

Area	Status
Base and FISCO contracts, protocol, bundler, member services, identity, TEEs on the simulator, agent with MCP, api, standing orders and payouts, audit and camt.053	Built and tested; gates M1 to M11 passed
WeCross transport	Built and tested (M12, gate passed: integration 16/16 files and e2e 41/41 on a fresh fork with `BRIDGE=wecross`, F9's freeze delivered through WeCross)
Web app with the consortium console, payees, devices, recovery page, web push, interim sandbox	Built and tested (M13, gate passed: the Playwright journey F1 to F10 and the sandbox spec 15/15, 34 screenshots at 1440 and 390 px, each with axe)
WeChat mini program	Built (M14, gate passed: logic tests, MANUAL steps written); on 2026-10-18 the team passed on an iPhone the link, a SOTER Face ID approval, a replayed approval refused on-chain, the regulator freeze, the unlink and the Chinese UI (MANUAL.md section 5); the Android fingerprint runs in Phase C
Agent key rotation, devices and owner recovery (F10)	Built and tested (M15); gate passed on fresh forks: `test/e2e/F10-rotation-recovery.test.ts` 3/3 (stop, rotate, recovery scheduled and cancelled) and the 48 h `test/warp/F10-recovery-execute.test.ts` (`executeRecovery`: only the new passkey signs in, the agent key is revoked)
Guomi (SM) profile smoke	Run and recorded in SMOKE.md: 11 self-test and 37 on-chain checks pass; WeIdentity scenario 39 of 43 (BLOCKERS B3)

Area	Status
Base Sepolia deployment and live mode	Contracts deployed on Base Sepolia and the consortium on FISCO (D74, D75), and the Base contracts deployed again on 2026-10-19 for the server (MandateHook `0x6411cc907Ace53F1Bdf8607a85991d3b38E79a5f`, D101; the M16 addresses stay in `infra/deployments/base-sepolia-m16.json`); real XAU and XAG prices from Pyth Hermes on live only, one request every 5 minutes (D89; the fork keeps the seeded demo prices its tests set); the live evidence run passed the M16 gate on 2026-10-18 05:35 UTC (D96): C0 to F8 on one account at `DEMO_SCALE=100` with Pyth prices, the real model and WeCross, 10 flows (F6 twice: its first attempt failed on the old agent instructions, issue #4, and was re-run on the same account after D93) with 21 BaseScan links to Base transactions and the FISCO calls in EVIDENCE.md; live reads Base through publicnode (D92)
Red-team suite	Built and run against the local stack on 2026-10-18, and again on 2026-10-19 as the final run before the public server, with the same counts (security/redteam.md): scripted 85 attacks, 0 executed; global (`make redteam-global`, shared state restored) 3 attacks, 0 executed; real model (DeepSeek through OpenCode) 12 attacks, 0 executed (3 blocked by the stack, 9 declined by the model; their scripted twins show the stack blocks the same attacks). The findings of the earlier runs, none of which moved value, are fixed and re-tested (D86 to D88). Held-out cases run through `REDTEAM_HELDOUT_DIR` (D64) and have not been run yet. Part of the M16 gate, passed on 2026-10-18 with the live evidence run (D96)
Production compose, rehearsal target, prod refusal check, phone tunnel	Built and rehearsed (M18, gate passed, D90): `make rehearsal` on the fork with `compose.prod.yml`, smoke F1, F5, F8 16/16 (scripted model) and the five prod refusals. Not rehearsed: live Base, Phala TEEs, a real certificate; all three are in the server deployment
Public server and TEEs on Phala Cloud	Deployed on 2026-10-20 (D106): https://app.praetorwallet.com on the Base deployment of D101 with its own FISCO consortium, bootstrapped with Pyth prices; agent-signer and soter-attester in Phala Cloud CVMs (Intel TDX), the attester activated on Base after the bank verified its quote; a WeChat link, a payment refused UNKNOWN_PAYEE, a SOTER approval through the attester CVM and the payment executed once, on an iPhone (EVIDENCE-SERVER.md)

The WeBASE explorer was proven in Phase 0 (V9) but is not part of the stack (cut list, D4).